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B. E. (Third Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(Mech., Mechatronics Engg. Branch)

FLUID MECHANICS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Part (a) of each question is compulsory and carrying 2 marks each and attempt two parts from (b), (c) and (d) carrying 7 marks each.

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(a) Differentiate between Newtonion and Non-Newtonian fluids. Give one example of each. 2

(b) Two horizontal plates 1·25 can apart, the space between them being filled with oil of viscosity 14 paises. Calculate shear stress in oil if upper plate is moved with a velocity of 2·5 m/s. 7

(c) A rectangular plane surface 2 m wide and 3 m deep lies in water in such a way that its plane makes an angle of 30° with the free surface of water. Determine the total pressure and position of centre of pressure when the upper edge is 1.5 m below, the free water surface. 7

(d) Find the volume of water displaced and position of centre of buoyancy for a wooden block of width 2·5 m and of depth 1·5 m, when it floats horizontally in water. The density of wooden block is 650 kg/m^2 and its length 6·0 m. 7

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2. (a) Explain steady and wasteady flow. 2

(b) A 25 cm diameter pipe carries oil of specific gravity 0.9 at a velocity of 3 m/s. At another section diameter is 20 cm. Find the velocity at this section and also mass rate of flow of oil. 7

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(c) A fluid flow field is given by

$$V = x^2 y\, i + y^2 z j - \left(2xyz + yz^2\right) k$$

Prove that it is a case of possible steady incompressible fluid flow. Calculate the velocity and acceleration at the point (2, 1, 3).

(d) The stream function for a two-dimensional flow is given by $\psi = 2xy$, calculate the velocity at point (2, 3). Find the velocity potential function ϕ. 7

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3. (a) What are the merits and demerits of orifice meter. 2

(b) State and derive Bernoulli's theorem. 7

(c) A horizontal venturimeter with inlet diameter 20 cm and throat diameter 10 cm is used to measure the flow of oil of specific gravity 0·8. The discharge of oil through venturimeter is 60 litre/second. Find the reading of the oil-mercury differential manometer Take $C_d = 0 \cdot 98$. 7

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(d) A 300 mm diameter pipe carries water under a head of 20 metre with a velocity of 3·5 m/s. If the axis of the pipe turns through 45°. Find the magnitude and direction of resultant force at the bend. 7

4. (a) Explain Reynold's Experiment. 2

(b) Derive Hagen-Poiseuille's equation. 7

(c) Determine the wall shealing stress in a pipe of diameter 100 mm which carries water. The velocities at the pipe centre and 30 mm from the pipe centre

are 2 m/s and 1·5 m/s respectively. The flow in pipe is given as turbulent. 7

(d) An oil of specific gravity 0·9 and viscosity 0.06 paise is flowing through a pipe of diameter 200 mm at the rate of 60 litre/s. Find the lead cost due to friction for a 500 m length of pipe. Find the power required to maintain this flow. 7

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5. (a) Explain the term dimensionally homogeneus equation. 2

(b) Write down the mathematical expression and application of the dimensionless numbers i.e Reynold's number. Froud's number, Euler number, Weber number and Mach number. 7

(c) The time period (t) of a pendulum depends upon the length (L) of the pendulum and acceleration due to gravity (g). Derive an expression for the time period. 7

(d) Derive on the basis of dimensional analysis suitable parameters to present the thrust developed by a propeller. Assume that the trust P depends upon the angular velocity w. spead V, diameter N dynamic viscosity μ, mass density ρ, elasticity of the fluid medium which can be developed by the speed of sound in the medium C. 7